CSCE 763: Digital Image Processing
Homework #1

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.
Assumptions:

(a) The distance between the center of the lens and the retina along the visual axis is 14 mm.

(b) The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.

(c) The fovea can be modeled as a square array of dimensions 1.5 mm × 1.5 mm, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Step 1: Cone Diameter on the Fovea

The fovea is modeled as a 1.5 mm × 1.5 mm square with 337,000 cones uniformly distributed.
If the cones are arranged in a square grid, the number of cones per side is:

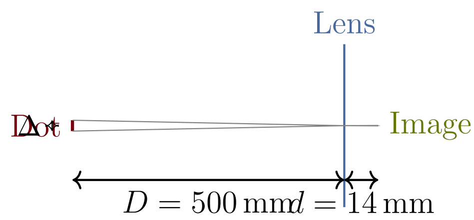
$$n = \sqrt{337{,}000} \approx 580.5$$

The pitch (center-to-center spacing, which equals the cone diameter including spacing) is:

$$d_{\text{cone}} = \frac{1.5\,\text{mm}}{580.5} \approx 0.00258\,\text{mm} = 2.58\,\mu\text{m}$$

Step 2: Similar Triangles

The dot on the page and its image on the retina are related by similar triangles through the lens:



By similar triangles, the relationship between the dot diameter Δ on the page and the image size d_{image} on the retina is:

$$\frac{\Delta}{D} = \frac{d_{\text{image}}}{d}$$

where $D = 500\,\text{mm}$ (distance to page) and $d = 14\,\text{mm}$ (lens-to-retina distance).

Step 3: Solve for Minimum Dot Diameter

The smallest detectable image equals one cone diameter: $d_{\text{image}} = d_{\text{cone}} \approx 0.00258\,\text{mm}$. Solving for Δ:

$$\Delta = D \cdot \frac{d_{\text{image}}}{d} = 500 \cdot \frac{0.00258}{14}$$

$$\Delta = \frac{500 \times 0.00258}{14} \approx 0.0923\,\text{mm}$$

Results

The smallest printed dot diameter that the eye can discern at a distance of $0.5\,\text{m}$ is:

$$\boxed{\Delta \approx 0.0922\,\text{mm} \approx 92.2\,\mu\text{m}}$$

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

Setup: Identify the Image and Subsets

The full 5×10 image grid (columns 0–9, rows 0–4). S_1 encloses columns 1–4, rows 0–3 (dashed). S_2 encloses columns 5–8, rows 0–3 (dashed). Row 4 and the outermost columns (0, 9) lie outside both subsets.

	S_1				S_2				
0	0	0	0	0	0	0	1	1	0
1	0	0	1	1	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	0	0	0	0	0	0

0,0,1,1,1,0,0,1,1,1

Grid values (row, col):

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	1	1
1	1	0	0	1	**1**	0	1	0	0
2	1	0	0	1	0	1	1	0	0
3	0	0	1	1	**0**	0	0	0	0

(The col-0 and col-9 values and row-4 values lie outside S_1 and S_2.)

S_1 pixels with value 1 (cols 1–4, rows 0–3): $(1,3) = 1$, $(1,4) = 1$, $(2,3) = 1$, $(3,2) = 1$, $(3,3) = 1$
S_2 pixels with value 1 (cols 5–8, rows 0–3): $(0,7) = 1$, $(0,8) = 1$, $(1,6) = 1$, $(2,5) = 1$, $(2,6) = 1$
The boundary between S_1 and S_2 falls between columns 4 and 5. The critical boundary pixels are:

- S_1 rightmost 1-valued: $(1,4) = 1$
- S_2 leftmost 1-valued: $(2,5) = 1$

These two pixels are **diagonal neighbors** (differ by 1 in both row and column).

Definitions Review

For two subsets S_1 and S_2, they are:

- **4-adjacent:** if some pixel $p \in S_1$ with value in V has a pixel $q \in S_2$ with value in V in its 4-neighborhood $N_4(p)$ (up, down, left, right).
- **8-adjacent:** if some pixel $p \in S_1$ with value in V has a pixel $q \in S_2$ with value in V in its 8-neighborhood $N_8(p)$ (including diagonals).
- **m-adjacent:** (mixed adjacency) p and q have values in V, and either (i) $q \in N_4(p)$, or (ii) $q \in N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels with values in V.

Part (a): 4-adjacency

Check whether any pixel in S_1 with value 1 has a **4-neighbor** in S_2 with value 1. The only S_1 pixel adjacent to S_2 with value 1 is $(1,4) = 1$. Its 4-neighbors in S_2:

- Right: $(1,5) = 0 \notin V$

No other S_1 pixel with value 1 sits on column 4. All other 1-valued S_1 pixels are at columns 2–3, too far from S_2.
S_1 **and** S_2 **are NOT 4-adjacent.**

Part (b): 8-adjacency

Check whether any pixel in S_1 with value 1 has an **8-neighbor** in S_2 with value 1. Consider $p = (1,4) = 1 \in S_1$. Its 8-neighbors in S_2:

- $(0,5) = 0$
- $(1,5) = 0$
- $(2,5) = 1 \in V$ ✓

The pixel $q = (2,5) = 1 \in S_2$ is a diagonal (8-)neighbor of $p = (1,4) = 1 \in S_1$, and both have values in $V = \{1\}$.
S_1 **and** S_2 **ARE 8-adjacent.**

Part (c): m-adjacency

For the diagonal pair $p = (1, 4)$ and $q = (2, 5)$, m-adjacency requires that $N_4(p) \cap N_4(q)$ contains **no** pixels with values in V.

$$N_4(p) = N_4(1, 4) = \{(0, 4),\ (2, 4),\ (1, 3),\ (1, 5)\}$$
$$N_4(q) = N_4(2, 5) = \{(1, 5),\ (3, 5),\ (2, 4),\ (2, 6)\}$$
$$N_4(p) \cap N_4(q) = \{(1, 5),\ (2, 4)\}$$

Check values:

- $(1, 5) = 0 \notin V$
- $(2, 4) = 0 \notin V$

No pixel in $N_4(p) \cap N_4(q)$ has a value in V, so the diagonal connection is **allowed** under m-adjacency.
S_1 **and** S_2 **ARE m-adjacent.**

Results

For $V = \{1\}$:

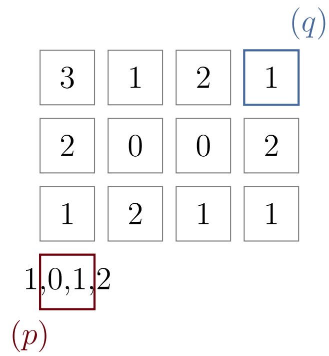
(a) **4-adjacent: No.** The nearest cross-boundary pixel pair with value 1 — $(1, 4)$ in S_1 and $(2, 5)$ in S_2 — are diagonal, not 4-neighbors.

(b) **8-adjacent: Yes.** Pixel $(1, 4) = 1 \in S_1$ and $(2, 5) = 1 \in S_2$ are 8-neighbors with values in V.

(c) **m-adjacent: Yes.** The diagonal pair $(1, 4)$ and $(2, 5)$ satisfies the m-adjacency condition because their shared 4-neighbors $\{(1, 5), (2, 4)\}$ both have value 0, which is $\notin V$.

Problem 3: Shortest Paths (25 pts)

Consider the image segment shown below.



The grid (using row, column indexing from top-left):

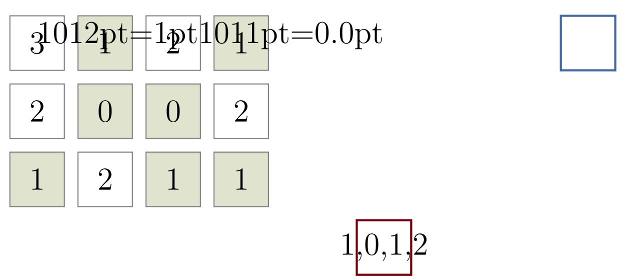
	col 0	col 1	col 2	col 3
row 0	3	1	2	**1 (q)**
row 1	2	0	0	2
row 2	1	2	1	1
row 3	**1 (p)**	0	1	2

$p = (3, 0)$ with value 1, $\quad q = (0, 3)$ with value 1.

(a) $V = \{0, 1\}$

Identify Valid Pixels

Pixels with values in $V = \{0, 1\}$:



Valid pixels (shaded): $(0, 1) = 1$, $(0, 3) = 1$, $(1, 1) = 0$, $(1, 2) = 0$, $(2, 0) = 1$, $(2, 2) = 1$, $(2, 3) = 1$, $(3, 0) = 1$, $(3, 1) = 0$, $(3, 2) = 1$.

Shortest 4-path

A 4-path uses only horizontal and vertical steps between pixels whose values are both in V.
One such path from $p = (3, 0)$ to $q = (0, 3)$:

$$(3, 0) \to (3, 1) \to (3, 2) \to (2, 2) \to (2, 3) \to (0, 3)$$

Wait — $(2, 3) = 1$ to $(0, 3) = 1$: we need $(1, 3) = 2 \notin V$. So we cannot go directly up from $(2, 3)$.
Let me find a valid 4-path. From $(2, 3) = 1$, the 4-neighbors with values in V: $(2, 2) = 1$, $(3, 3) = 2$ (no). Going up: $(1, 3) = 2$ (no). So $(2, 3)$ is a dead end going upward.
Try via top: $(3, 0) \to (2, 0) \to \ldots$ $(2, 0) = 1$. 4-neighbors in V: $(3, 0) = 1$ (back), $(1, 0) = 2$ (no). Dead end.
Another attempt: $(3, 0) \to (3, 1) \to (3, 2) \to (2, 2) \to (1, 2) \to (1, 1) \to (0, 1) \to (0, 3)$? Check: $(0, 1) = 1$ to $(0, 3) = 1$: need $(0, 2) = 2 \notin V$. Cannot step from $(0, 1)$ to $(0, 3)$ directly — they are not 4-neighbors.
$(0, 1) = 1$: 4-neighbors in V: $(1, 1) = 0$, $(0, 0) = 3$ (no), $(0, 2) = 2$ (no). Dead end going right.
No 4-path exists from p to q, because pixel $(0, 1)$ cannot connect rightward to $(0, 3)$ — the intervening pixel $(0, 2) = 2 \notin V$, and there is no other route to reach column 3 in row 0.

Shortest 8-path

An 8-path allows diagonal moves. From the valid pixel map:
$(3,0) \to (3,1) \to (3,2) \to (2,3) \to (0,3)$? Check $(2,3) = 1$ to $(0,3)$: not 8-neighbors (distance = 2 rows). Invalid.
$(3,0) \to (3,1) \to (2,2) \to (1,1) \to (0,1)$: need diagonal. $(3,1) = 0$ to $(2,2) = 1$: 8-neighbor, both in V. $(2,2) = 1$ to $(1,1) = 0$: 8-neighbor, both in V. $(1,1) = 0$ to $(0,1)$: wait, is there a diagonal path onward?
$(0,1) = 1$. 8-neighbors in V: $(1,1) = 0$, $(1,2) = 0$, $(0,0) = 3$ (no), $(0,2) = 2$ (no). Still stuck.
Try: $(1,2) = 0$ to $(0,3) = 1$: these are 8-neighbors! Both in V.
Path: $(3,0) \to (3,1) \to (2,2) \to (1,2) \to (0,3)$
Check each step:

- $(3,0) = 1 \to (3,1) = 0$: 4-neighbor, both in V. ✓
- $(3,1) = 0 \to (2,2) = 1$: 8-neighbor (diagonal), both in V. ✓
- $(2,2) = 1 \to (1,2) = 0$: 4-neighbor, both in V. ✓
- $(1,2) = 0 \to (0,3) = 1$: 8-neighbor (diagonal), both in V. ✓

Length = 4 steps. This is optimal since the minimum Chebyshev distance from $(3,0)$ to $(0,3)$ is $\max(3,3) = 3$, so the shortest 8-path has length ≥ 3. Can we find a length-3 path?
A length-3 path requires 3 diagonal steps: $(3,0) \to (2,1) \to (1,2) \to (0,3)$. Check: $(2,1) = 2 \notin V$. Invalid.
Shortest 8-path length = 4.

Shortest m-path

An m-path uses mixed adjacency: diagonal steps are allowed only when the two pixels share no common 4-neighbor with a value in V.
Check the 8-path $(3,0) \to (3,1) \to (2,2) \to (1,2) \to (0,3)$:
Step $(3,1) \to (2,2)$ (diagonal): Common 4-neighbors of $(3,1)$ and $(2,2)$ are $(3,2)$ and $(2,1)$. $(3,2) = 1 \in V$. Since there is a common 4-neighbor in V, this diagonal move is **not allowed** in m-adjacency.
Alternative: replace the diagonal with a 4-connected detour via $(3,2)$: $(3,0) \to (3,1) \to (3,2) \to (2,2) \to (1,2) \to (0,3)$
Check $(1,2) \to (0,3)$ (diagonal): Common 4-neighbors of $(1,2)$ and $(0,3)$ are $(0,2)$ and $(1,3)$. $(0,2) = 2 \notin V$, $(1,3) = 2 \notin V$. No common 4-neighbor in V, so this diagonal is **allowed**.
Full m-path: $(3,0) \to (3,1) \to (3,2) \to (2,2) \to (1,2) \to (0,3)$
Shortest m-path length = 5.

Results

For $V = \{0, 1\}$:

Path Type	**Shortest Length**
4-path	**Does not exist** (no connected 4-path from p to q)
8-path	$\boxed{4}$
m-path	$\boxed{5}$

Problem 3 — Part (b): $V = \{1, 2\}$

Identify Valid Pixels

Pixels with values in $V = \{1, 2\}$:

	col 0	col 1	col 2	col 3
row 0	3	**1**	**2**	**1 (q)**
row 1	**2**	0	0	**2**
row 2	**1**	**2**	**1**	**1**
row 3	**1 (p)**	0	**1**	**2**

Shortest 4-path

$(3, 0) \to (2, 0) \to (2, 1) \to (2, 2) \to (2, 3) \to (1, 3) \to (0, 3)$
Check: all values in V and all steps are 4-neighbors.

- $(3, 0) = 1$, $(2, 0) = 1$, $(2, 1) = 2$, $(2, 2) = 1$, $(2, 3) = 1$, $(1, 3) = 2$, $(0, 3) = 1$ — all $\in V$. ✓

Length = 6. Can we do better? The Manhattan distance is $|3 - 0| + |0 - 3| = 6$, so this is optimal.

Shortest 8-path

The Chebyshev distance is $\max(3,3) = 3$, so the minimum possible 8-path length is 3.
Try: $(3,0) \to (2,1) \to (1,2) \to (0,3)$ Check: $(2,1) = 2 \in V$, $(1,2) = 0 \notin V$. Invalid.
Try: $(3,0) \to (2,1) \to (1,0) \to (0,1) \to (0,2) \to (0,3)$? $(1,0) = 2 \in V$, $(0,1)$: is $(1,0)$ 8-adjacent to $(0,1)$? Yes (diagonal). $(0,1) = 1 \in V$, $(0,2) = 2 \in V$, $(0,3) = 1 \in V$. Length = 5.
Shorter: $(3,0) \to (2,1) \to (1,2)$: $(1,2) = 0 \notin V$. No.
$(3,0) \to (2,0) \to (1,0) \to (0,1) \to (0,2) \to (0,3)$. Length = 5. All in V: $1,1,2,1,2,1$. ✓
Try length 4: $(3,0) \to (2,1) \to (1,0) \to (0,1) \to (0,3)$? $(0,1)$ and $(0,3)$ are not 8-neighbors. No.
$(3,0) \to (2,1) \to (1,2)$: invalid (value 0).
$(3,0) \to (2,0) \to (1,0) \to (0,1) \to (0,2) \to (0,3)$: length 5.
Try: $(3,0) \to (2,1) \to (1,0) \to (0,1) \to (0,2) \to (0,3)$: length 5.
Try length 4: $(3,0) \to (2,1) \to (1,0) \to (0,1) \to (0,3)$? Not adjacent. No.
$(3,0) \to (2,0) \to (1,0) \to (0,1) \to (0,3)$? $(0,1)$ to $(0,3)$: distance 2. No.
Try: $(3,0) \to (2,1) \to (1,2)$: value 0. Dead end.
Try via right side: $(3,0) \to (3,2) \to (2,3) \to (1,3) \to (0,3)$? $(3,0)$ to $(3,2)$: not 8-neighbors (distance 2). No.
$(3,0) \to (2,1) \to (2,2) \to (1,3) \to (0,3)$: length 4. Check: $(2,1) = 2$, $(2,2) = 1$, $(1,3) = 2$, $(0,3) = 1$. All $\in V$. All 8-neighbors? $(3,0) \to (2,1)$: diagonal ✓. $(2,1) \to (2,2)$: 4-neighbor ✓. $(2,2) \to (1,3)$: diagonal ✓. $(1,3) \to (0,3)$: 4-neighbor ✓.
Shortest 8-path length = 4.

Shortest m-path

Check the 8-path $(3,0) \to (2,1) \to (2,2) \to (1,3) \to (0,3)$:
Step $(3,0) \to (2,1)$ (diagonal): Common 4-neighbors: $(2,0)$ and $(3,1)$. $(2,0) = 1 \in V$. Diagonal not allowed in m-adjacency.
Step $(2,2) \to (1,3)$ (diagonal): Common 4-neighbors: $(2,3)$ and $(1,2)$. $(2,3) = 1 \in V$. Diagonal not allowed.
So this particular path is not valid for m-adjacency. We need to replace disallowed diagonals with 4-connected detours.
Optimal m-path: $(3,0) \to (2,0) \to (2,1) \to (2,2) \to (2,3) \to (1,3) \to (0,3)$
All 4-connected steps with values in V. **Length = 6.**
Can we do better? Try mixing: $(3,0) \to (2,0) \to (2,1) \to (2,2) \to (1,3) \to (0,3)$. Length 5. Check $(2,2) \to (1,3)$: diagonal. Common 4-neighbors: $(1,2) = 0 \notin V$, $(2,3) = 1 \in V$. Not allowed.
$(3,0) \to (2,0) \to (2,1) \to (1,2)$: $(1,2) = 0 \notin V$. Invalid.
Try: $(3,0) \to (2,1) \to ...$: $(3,0) \to (2,1)$ diagonal. Common 4-neighbors of $(3,0)$ and $(2,1)$: $(3,1) = 0$ and $(2,0) = 1 \in V$. Not allowed.
Shortest m-path length = 6.

Results

For $V = \{1, 2\}$:

Path Type	**Shortest Length**
4-path	$\boxed{6}$
8-path	$\boxed{4}$
m-path	$\boxed{6}$

Problem 4: Set Expressions (25 pts)

Give expressions for the sets shown shaded in the following figure in terms of sets A, B, and C.

In the figure: A is a rectangle (top), B is a triangle (bottom-left), C is a triangle (bottom-right). The three shapes overlap.

Analyzing Each Figure

Figure (i): The shaded region is the part of A that overlaps with B but not C, plus the part of A that overlaps with C but not B. This is the part of A that is in exactly one of B or C:

$$\boxed{(A \cap B \cap C^c) \cup (A \cap C \cap B^c)}$$

Equivalently: $A \cap (B \oplus C)$ where $\oplus$ denotes symmetric difference, or simply $A \cap (B \cup C) \cap (B \cap C)^c$.

Figure (ii): The shaded region includes the part of B not in A or C, plus the intersection of A and C (including overlaps with B). Looking at the shading pattern:

$$\boxed{(A \cap C) \cup (B \cap A^c \cap C^c)}$$

Figure (iii): The shaded region is the part of B and C that is not in A. That is, the union of B and C minus A:

$$\boxed{(B \cup C) \cap A^c}$$

Results

(i) $(A \cap B \cap C^c) \cup (A \cap C \cap B^c)$

(ii) $(A \cap C) \cup (B \cap A^c \cap C^c)$

(iii) $(B \cup C) \cap A^c$